CSE 4095

Special Topics in Computer Engineering III: Introduction to Embedded Systems

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About the Instructor

- Dr. Sanem Arslan Yılmaz
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  - Office: M2-230
  - Office Hours: Tuesday 14:00 - 16:00
    Thursday 10:30 - 12:30 or by appointment

- Lectures: Tuesday 09:30 – 10:30
  Thursday 15:00 – 17:00

Course Organization

- Course materials, assignments, and submissions will be held through Canvas.
  - Students can self-enroll in the course using this URL:
    - **https://canvas.instructure.com/enroll/ARDTDR**
    - Alternatively, they can sign up at **https://canvas.instructure.com/register** and use the following join code: **ARDTDR**
  - Please set your time-zone on Canvas (Istanbul GMT+3)
  - It is **YOUR RESPONSIBILITY** to make sure that your account receives the messages properly. We will send the announcements through Canvas and assume you read it the same day.

Course Organization (cont.)

- Reference Books:
  1. Yifeng Zhu: **Embedded Systems with ARM Cortex-M Microcontrollers in Assembly Language and C,** Publisher: E-Man Press LLC; 3rd edition (2017)
  2. Lee and Seshia: **Introduction to Embedded Systems - A Cyber-Physical Systems Approach**, MIT Press, Second Edition (2017).
  3. Peter Marwedel: **Embedded System Design**, Springer, 2006.
  4. Jonathan Valvano, **Embedded Systems: Real-Time Operating Systems for ARM Cortex-Microcontrollers**, Volume 3, fourth edition, January 2017
  5. Marilyn Wolf : **Computers as Components, Principles of Embedded Computing System Design**, Mourgan Kaufmann Publishers, 2nd edition.
- The slides contain ideas/materials from reference books.
- Grading:
  - Quiz & Homework & Projects **30%**
  - Midterm **30%**
  - Final **40%**

Assignment Policy

- **Programming Projects:** There will be tentatively 2 programming project(s) done by groups of 2-3 people (probably using simulators like *ARM Keil*).
- **Homework/Exercise:** There might be handwritten homework assignments (probably done individually).
- You are encouraged to discuss an assignment and the general approach to a problem with your instructor or your classmates.
- However, the final submitted work has to be totally yours. You have to design and write the code yourself.
- Plagiarism is a very serious offense in the academic world. All types of plagiarism will cause you to take an **FF** grade from the course!
- No late submission will be accepted.

About this course

- Theoretical foundations and principles of the analysis and design of embedded systems.

- Practical aspects of embedded system design, mainly software design.

Embedded Systems

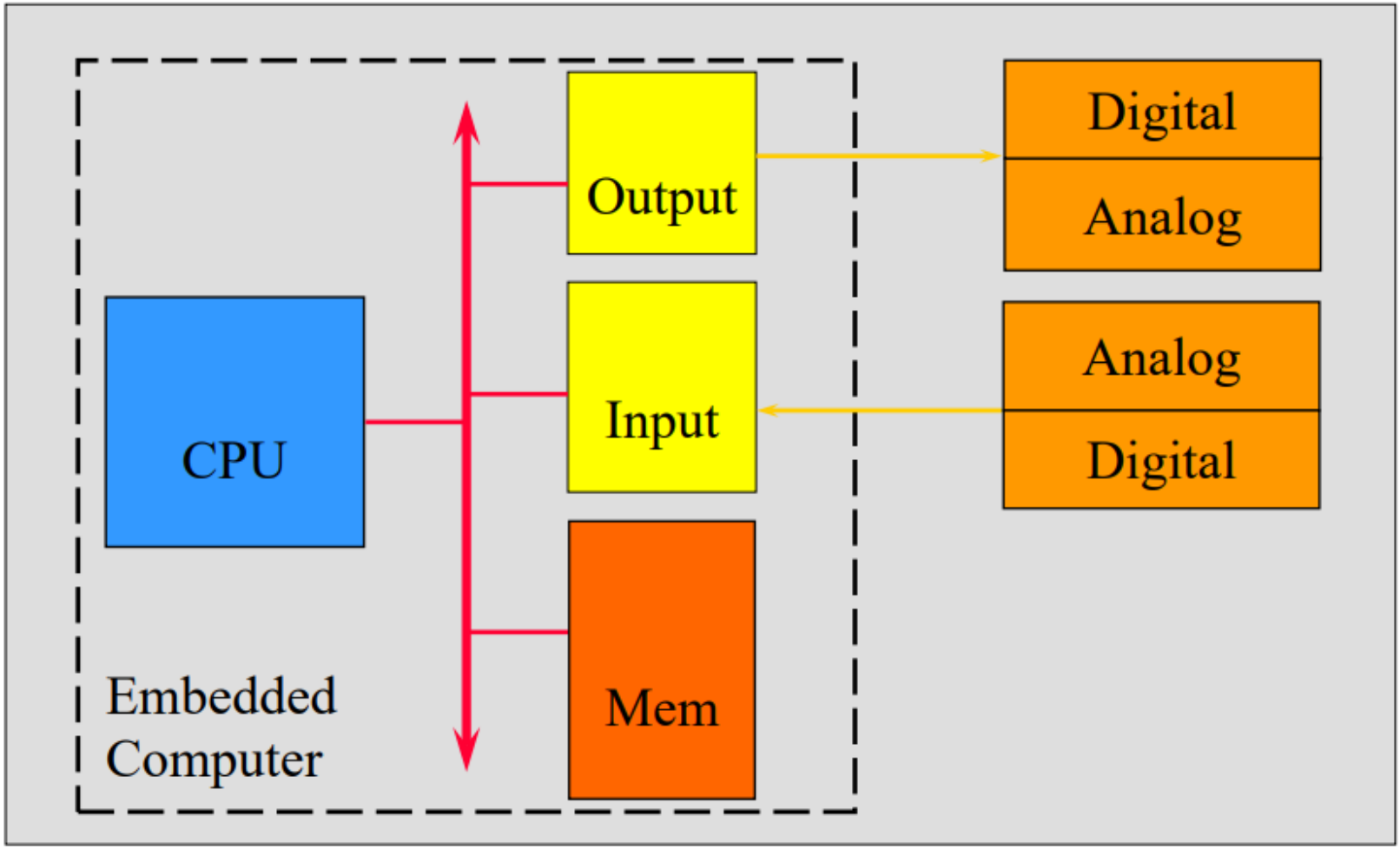
- Embedded systems (ES) = information processing systems embedded into a larger product
  - Automobiles, cell phones, household appliances,..
- Connected to the physical environment through ***sensors*** collecting information about that environment and ***actuators*** controlling that environment

Examples:



Often, the main reason for buying is not information processing

Embedded Systems



Embedded Systems in Automobiles

- Airbag: commands from a microcontroller
- Navigation: GPS system
- Adaptive cruise control: safety distance from other vehicles
- Anti-braking systems (ABS)
- Blind-angle alert systems
- Rain-sensing: optical sensor
- Automatic parking system: sensors as both a transmitter and receiver
- Self-driving car
  - a vehicle that is capable of sensing its environment and moving with little or no human input
  - need to understand the safest way to react to pedestrians, other cars on the road, road obstructions, and other driving risks
  - combination of different embedded systems of today

Embedded Systems in Avionics

- Flight control systems,
- Anticollision systems,
- Pilot information systems,
- Autopilots, etc.
- Autonomous flying is also becoming a reality (at least for certain application areas)

Dependability is of utmost importance.

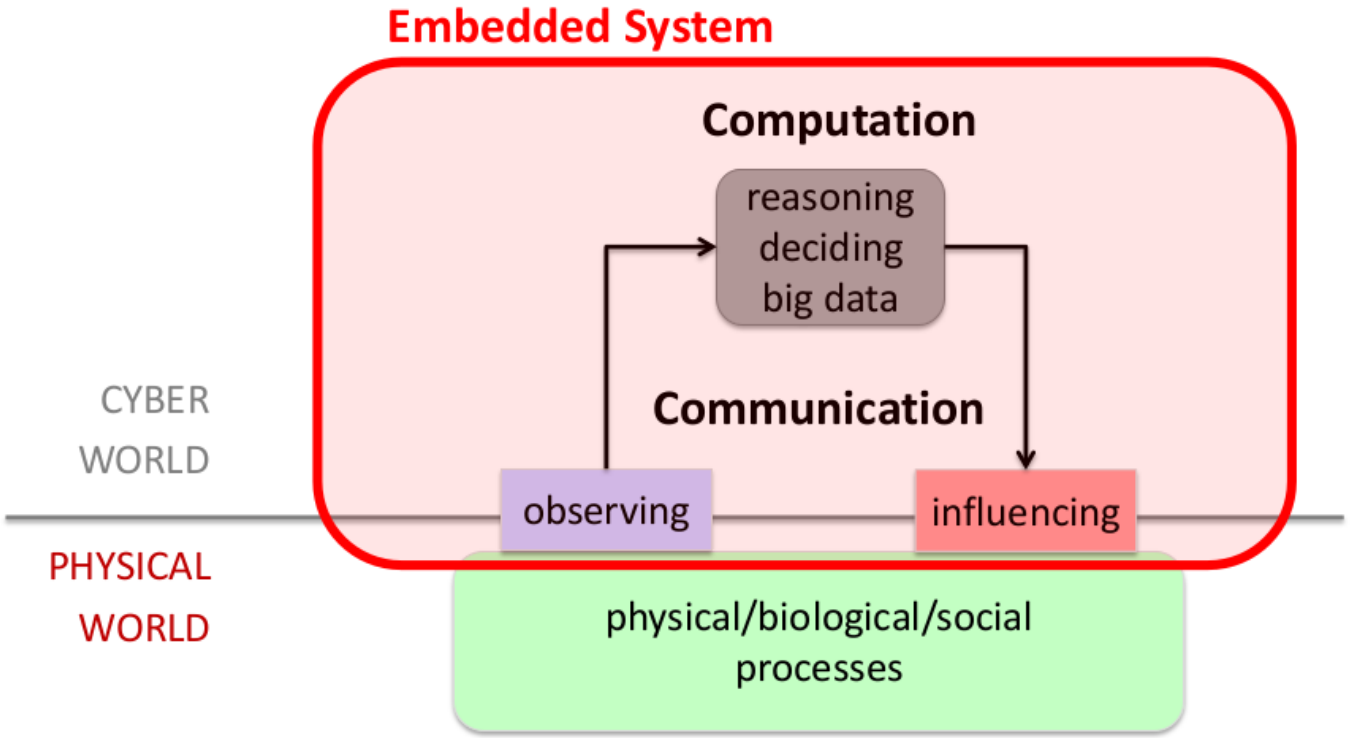
Embedded Systems in Smart Homes

- IP multimedia door entry system: view a caller in the entrance, have control of their access
- IP television
- In-home audio and video streaming
- Remote health monitoring

And many more..

- Medical systems, telecommunication, robotics, military applications, smart buildings, consumer electronics, etc.
- Almost all engineering disciplines will be affected.

Embedded System



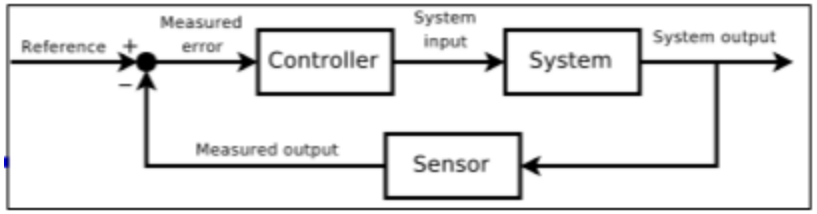
Use feedback to influence the dynamics of the physical world by taking smart decisions in the cyber world

Main Characteristics

- Dependable
  - Reliability, maintainability, availability, safety, security
  - e.g. airplanes are safety-critical



- Predictable
  - How a system is going to behave under any circumstances

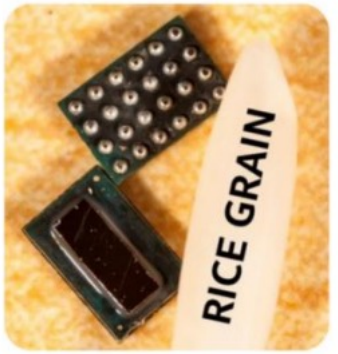


Main Characteristics (cont.)

- Timing
  - Embedded systems often must meet **real-time** constraints.
  - Have to perform in real-time
    - If the data is not ready by a certain deadline, the system breaks.
  - For hard real-time systems, right answers arriving too late are wrong. All other time-constraints are called soft.
- Hybrid systems
  - Include analog and digital parts
- Reactive systems
  - Be in a certain state, waiting for an input, perform some computation for each input, and generate an output and a new state

Main Characteristics (cont.)

- Efficiency
  - Embedded systems must be efficient.
    - Energy efficient
    - Code-size and data memory efficient
    - Run-time efficient
    - Weight efficient
    - Cost efficient



- Specialized
  - Embedded Systems are often specialized towards a certain application or application domain.
- Dedicated user interface
  - Not regular keyboards, mouse, or monitor
  - May contain buttons, steering wheels, pedals, sensors, etc.

Comparison

- **Embedded Systems**
  - Dedicated for a certain application
  - Not programmable by end user
  - Fixed run-time requirements (faster may not be essential)
- Criteria
  - Cost
  - Power consumption
  - Size and weight
  - Dependability
  - Worst-case performance

- **General Purpose**
  - Variety of applications
  - Programmable by end user
  - Faster is better
- Criteria
  - Cost
  - Power consumption
  - Average speed

Course Content (Tentative)

- Introduction to Embedded Systems
- **Embedded Software:**
  - Coding for embedded systems
  - ARM Assembly
  - Mixing C with Assembly
  - Interrupts
  - GPIO
- **Embedded Operating Systems:**
  - WCET Analysis
  - Real-time Scheduling
  - Optimizations
- **Embedded Hardware:**
  - Embedded Processors, Parallelism
  - Memory Architectures
  - Energy Efficiency & Reliability